

HaluCheck: Explainable and verifiable automation for detecting hallucinations in LLM responses

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ABSTRACT

Large language models have become integral to various aspects of modern life, but a critical challenge persists: hallucinations. This work contributes to expert systems research by providing a systematic framework for enhancing AI reliability and decision support capabilities. This paper introduces **HaluCheck**, a visualization system that assesses and prominently displays the likelihood of hallucination in model responses. The system allows users to select from various evaluation methods, including an automated pipeline named **AutoFactNLI** that decomposes responses, retrieves relevant documents, and evaluates each sentence as potentially hallucinatory or not. By integrating user-provided API keys for multiple LLM models, our system facilitates horizontal comparisons of factual reliability and response quality across different models, emphasizing the importance of factual accuracy in high-stakes domains like finance, law, healthcare, and journalism. Comprehensive experimental results from HaluEval demonstrate that AutoFactNLI outperformed the LLM-as-a-Judge approach, where GPT-4 evaluates responses without explicit guidance or tailored instructions, relying solely on its inherent training to classify outputs as factual or hallucinated. HaluCheck underscores the value of a user-centered system that not only enables real-time factuality assessments but also fosters transparency and adaptability, making it a vital tool for improving trust in AI-driven decision support systems across critical domains.

1. Introduction

Large Language Models (LLMs) have become transformative tools, enhancing information discovery, productivity, and research across diverse domains. Despite their utility, a persistent challenge remains: hallucinations—responses that appear plausible but lack factual accuracy or contain inconsistencies (Honovich et al., 2022; Lee, Firat, Agarwal, Fannjiang, & Sussillo, 2018; Maynez, Narayan, Bohnet, & McDonald, 2020). These issues are particularly concerning in domain-specific applications where experts require precise, reliable information (Luo, Xie, & Ananiadou, 2024; Zeiser, da Costa, de Oliveira Ramos, Maier, & da Rosa Righi, 2024). While strategies like retrieval-augmented generation (RAG) (Chen, Pasupat, Singh, Lee, & Guu, 2023; Gao et al., 2022; Lewis et al., 2020) and citation augmentation (Li, Li, Ma, & Liu, 2024) provide grounding mechanisms, post-hoc evaluation systems have gained traction as essential tools for improving factual accuracy and decision support.

Although advanced evaluation methods such as SelfCheckGPT (Manakul, Liusie, & Gales, 2023b) demonstrate promising results by detecting contradictions through multiple output generation, they often lack precision in identifying granular errors and transparency in explaining their judgments. For experts in high-stakes domains such as healthcare, law, and finance, these limitations pose a significant barrier to trust and adoption. A reliable system must not only evaluate outputs but also align with the expectations and workflows of expert users, enabling them to scrutinize and refine responses effectively.

To address these needs, we introduce **HaluCheck**, a visualization system designed to provide expert users with a dynamic, flexible, and explainable platform for assessing and mitigating hallucinations in LLM responses. At its core, **HaluCheck** integrates **AutoFactNLI**, an automated hallucination detection pipeline that decomposes responses into atomic facts, evaluates their factuality against external

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knowledge sources, and visualizes areas of potential inaccuracy. Unlike methods that focus solely on holistic accuracy scores, **AutoFactNLI** empowers users to pinpoint and correct specific problematic segments within a response, offering granular error localization tailored to expert workflows.

Our system prioritizes expert usability and trustworthiness through the following key contributions:

Granular Error Localization and Correction: **AutoFactNLI** evaluates individual sentences and atomic facts, enabling the identification of specific inaccuracies within a response. This feature is particularly valuable for expert users who require precise assessments and actionable insights for refining outputs.

Explainability and Transparency: In contrast to LLM-based evaluation methods, which rely on opaque internal reasoning, **AutoFactNLI** grounds its assessments in external, verifiable knowledge sources. This approach ensures that users can understand and trust the rationale behind the factuality evaluations.

Flexibility Across Models and Metrics: **HaluCheck** enables users to compare multiple LLMs and evaluation methods dynamically. By integrating diverse metrics such as BERTScore (Zhang, Kishore, Wu, Weinberger, & Artzi, 2019) and entailment-based approaches, it allows expert users to tailor the evaluation process to their specific needs and preferences.

Our work contributes to the development of expert-oriented systems by addressing the practical challenges of hallucination detection and response refinement. By combining theoretical advancements in atomic fact decomposition with user-centric design, **HaluCheck** bridges the gap between research innovations and expert usability.

We validate our approach through extensive experimentation using the **HaluEval** (Li, Cheng, Zhao, Nie, & Wen, 2023) dataset, demonstrating the system's capability to provide actionable insights and enhance trustworthiness. With its focus on granular evaluation, explainability, and expert usability, **HaluCheck** and **AutoFactNLI** lay a robust foundation for AI applications in high-stakes fields demanding precision and accountability, such as journalism, law, finance, and healthcare.

2. Related works

Large Language Models (LLMs) often produce contextually plausible but hallucinated responses due to limitations in transformer architectures, biases in training data, and exposure bias (Brown et al., 2020; Ji et al., 2023; Rawte, Sheth, & Das, 2023; Touvron et al., 2023; Wang & Sennrich, 2020). Although various strategies (e.g., Retrieval-Augmented Generation) have been explored to reduce such hallucinations (Chen et al., 2023; Gao et al., 2022; Jiang et al., 2023; Lewis et al., 2020; Rebuffel et al., 2022), our focus is on evaluating model outputs at inference rather than mitigating hallucinations during training. Specifically, **HaluCheck** aims to measure hallucinations in generated text through a post-hoc evaluation pipeline that highlights unverified or questionable content.

This section is organized into two parts:

- **Hallucination Evaluation:** Summarizes methods for quantifying hallucinations.
- **Visualization Systems for Comparing Multiple LLMs:** Reviews tools enabling users to compare outputs and detect potential hallucinations across different models.

2.1. Hallucination evaluation

Evaluation methods vary, with some verifying consistency against independent reference data sources and others leveraging the LLM's generative and inferential capabilities without external resources. Min et al. introduced **FActScore** (Min et al., 2023) (Factual Precision in Atomicity Score) system, which breaks down generated text into atomic facts for verification against Wikipedia. This system, however, faces

challenges in achieving full automation due to the need for costly human labeling.

The **FActScore** system (Factual Precision in Atomicity Score) decomposes generated text into discrete atomic facts, verified individually against trusted knowledge sources. It uses **InstructGPT** (text-davinci-003) (Ouyang et al., 2022) to break down generated text into sentences and refine them into atomic facts. In an 8-shot learning context, **InstructGPT** is provided with eight examples demonstrating how to deconstruct complex sentences into atomic facts.

The system uses biographical information and compares it with information available on Wikipedia. Factual precision is calculated by assessing each atomic fact with equal importance. Human annotators classify each atomic fact as *supported*, *not-supported*, or *irrelevant* when compared to English Wikipedia.

The **ALCE** framework (Gao, Yen, Yu, & Chen, 2023) goes beyond mere citation generation to evaluate fluency, correctness, and citation quality but has limitations. Its effectiveness is constrained when applied to datasets without pre-determined answer lists, such as **ELI5** (Fan et al., 2019), requiring unique automated evaluation methods to assess correctness. The reliance on pre-existing short answer lists in datasets like **ASQA** and **QAMPARI** (Amouyal et al., 2022; Stelmakh, Luan, Dhingra, & Chang, 2022) limits its direct applicability to general QA contexts outside these datasets.

An example of such an automated evaluation method is the **ALCE** framework's approach to the **ELI5** dataset. Here, **InstructGPT** (Ouyang et al., 2022) generates sub-claims, and the responses are evaluated based on the number of sub-claims entailed, using a method known as claim recall. This approach underscores the need for adaptable evaluation strategies in the absence of predetermined answers, paving the way for more flexible and comprehensive QA assessments.

Recent advancements have gone further to regenerate answers by replacing non-factual segments with factual information based on generated content (Li et al., 2024). However, the lack of publicly available code for this research introduces limitations in reproducibility and verifiability.

Manakul et al. (2023b) provide an example of research utilizing the generative and inferential capabilities of the LLM model itself, without external resources. This study introduces **SelfCheckGPT**, a zero-resource method to detect hallucinations by evaluating consistency across multiple LLM outputs.

SelfCheckGPT operates on the principle that factual information will yield similar and consistent responses from an LLM, while hallucinated content will show significant variability and contradictions. By generating and comparing a series of responses to the original LLM response from a single query, it assesses the degree of alignment among these responses. Depending on the method used, it assigns a hallucination score to each sentence on a scale from 0.0 to 1.0, or uses other metrics like n-gram log-probabilities.

SelfCheckGPT employs five different methods for evaluating responses to identify hallucinations:

1. **BERTScore:** Computes the average BERTScore between the sentence in question and the most similar sentence from each response sample.
2. **Question-Answering (QA):** Utilizes the MQAG (Manakul, Liusie, & Gales, 2023a) framework to generate multiple-choice questions based on the primary response and assesses consistency through the answers from the sampled responses.
3. **n-gram Analysis:** Develops a simple language model from the samples and the main response to estimate the LLM's token probabilities. Consistency is evaluated based on the presence of n-grams within the generated samples.
4. **Natural Language Inference (NLI):** Uses a contradiction score from an NLI model to determine if the assessed sentence contradicts the content of sampled responses, indicating hallucination.
5. **Prompting:** Directly queries an LLM to evaluate whether the sentence is supported by the context provided in the samples, leveraging

the LLM's reasoning abilities to assess information consistency.

Among these methods, the NLI-based approach was evaluated as striking a good balance between performance and computational efficiency, making it a practical option for detecting hallucinations.

This fully automated evaluation technique, which does not require human annotators, is suitable for integration into **HaluCheck**, an automated hallucination evaluation system for general QA contexts. This method contrasts with approaches that involve human annotation in the evaluation process.

Recent advancements in hallucination detection for LLMs emphasize the development of scalable frameworks tailored to diverse contexts. Approaches such as the Fact-Level Calibration framework refine responses by self-correcting low-confidence segments (Yuan et al., 2024), while benchmarks like DAHL address domain-specific needs, particularly in biomedical contexts, by highlighting limitations in parameter scaling (Seo, Lim, Jang, & Shin, 2024). Zero-shot methods such as Hallucination Reasoning evaluate the integration of intrinsic knowledge without external datasets (Lee, Hsu, & Chen, 2024), and systems like SCAR introduced by Härle et al. (2024), allow for ethical and customizable output steering. Collectively, these strategies underscore a shift toward fine-grained, domain-adapted, and ethically driven solutions that enhance the reliability of LLM outputs across various applications.

2.2. Visualization systems for comparing multiple LLMs

There have been several efforts to develop visualization systems that enable the horizontal comparison of multiple LLMs. Kim, Lee, Shin, Kim, and Kim (2023) focus on the interactive evaluation of LLMs using user-defined criteria through a system called EvalLM. This system aims to assist prompt designers in refining their prompts by evaluating multiple outputs based on criteria specified by users in natural language. While this approach allows for a wide range of criteria to be set, it relies on the LLM for evaluation standards, which poses challenges in verifying the appropriateness of the LLM's evaluation metrics.

LLM Comparator (Kahng et al., 2024) is designed to evaluate and compare the quality of responses generated by multiple LLMs through automatic side-by-side evaluation. It enables users to analyze when, why, and how one model performs better or worse than a baseline model by providing interactive workflows and visualization tools.

This paper introduces an advancement in the domain of LLM evaluation by focusing on automating the hallucination evaluation process using an external database. The motivation to automate arises from the recognition that existing methods, which rely on human judgment to determine the truthfulness of a sentence or the presence of true answers within datasets, can be both time-consuming and costly. Automation presents an opportunity for broader applicability across general QA contexts and promises substantial cost reductions by minimizing the dependency on human labeling. A zero-resource and automated evaluation system like SelfCheckGPT exemplifies the feasibility and ease of integration of this approach, marking a critical step towards more efficient and scalable hallucination detection.

Despite the evident utility of such a system, the paper notes the scarcity of existing visualization tools tailored for this purpose, highlighting a gap in the current LLM evaluation toolkit.

3. System design

In this section, we provide an overview of the design and main functionalities of **HaluCheck**. We begin by describing the key components that form the core of the system, followed by a detailed explanation of the user interface layout and interaction. Finally, we illustrate the underlying data flow to show how user inputs and system modules integrate to produce real-time evaluation results.

3.1. Key functionalities

Effective visualization in conversational systems is crucial for understanding and improving interactions. Research has demonstrated that visualizations can enhance the user experience by providing clear, actionable insights into conversations. According to Kim, Calvo, Enfield, and Yacef (2021), visualizations allow users to explore and understand complex data, thereby improving the efficiency and effectiveness of conversational systems.

The development of **HaluCheck** is driven by the necessity to ensure the reliability and accuracy of outputs generated by LLMs. **HaluCheck** aims to address this challenge by providing a user-friendly interface and robust evaluation methods. This subsection explains the system's design and the essential functionalities required to achieve its objectives.

3.1.1. Model selection hub

The model selection hub enables users to choose from various model engines for response generation, allowing seamless switching between models and comparison of outputs under identical conditions. This flexibility is crucial for researchers and professionals needing to efficiently evaluate multiple models' performance.

3.1.2. User authentication and method selection

This function manages user authentication through API key input, ensuring secure and personalized usage. It offers various evaluation methods, including AutoFactNLI, SelfCheckNLI from SelfCheckGPT (Manakul et al., 2023b), and Prompt Evaluation, allowing users to choose the most suitable techniques for detecting hallucinations according to their specific needs.

3.1.3. Interaction and evaluation display

The interaction and evaluation display functionality enables users to input queries, receive model responses, and view detailed evaluation results. Sentences identified as hallucinated by the chosen evaluation method are distinctly highlighted, providing immediate visual feedback to help users identify and address inaccuracies. This sentence-level analysis allows users to verify the factual accuracy of highlighted sentences using other resources, offering actionable feedback crucial for improving the quality of LLM outputs and building trust in their applications.

By incorporating functionalities such as model selection, secure user access, and detailed evaluation displays, **HaluCheck** ensures that users can effectively evaluate and enhance the reliability of LLM applications.

3.2. User interface

HaluCheck's user interface (UI) is designed to provide a comprehensive and user-friendly platform for assessing the likelihood of hallucinations in LLM outputs. This design caters to the immediate needs of researchers and professionals while enhancing the overall reliability of LLM applications.

The UI is segmented into distinct areas — upper, left, and center panes — each serving specific functionalities to streamline the user experience.

3.2.1. Upper pane

The upper pane functions as the model selection hub, allowing users to select different model engines such as gpt-3.5-turbo or gpt-4o. This provides flexibility for evaluating various LLMs under identical conditions, enabling efficient comparison and analysis of model performance.

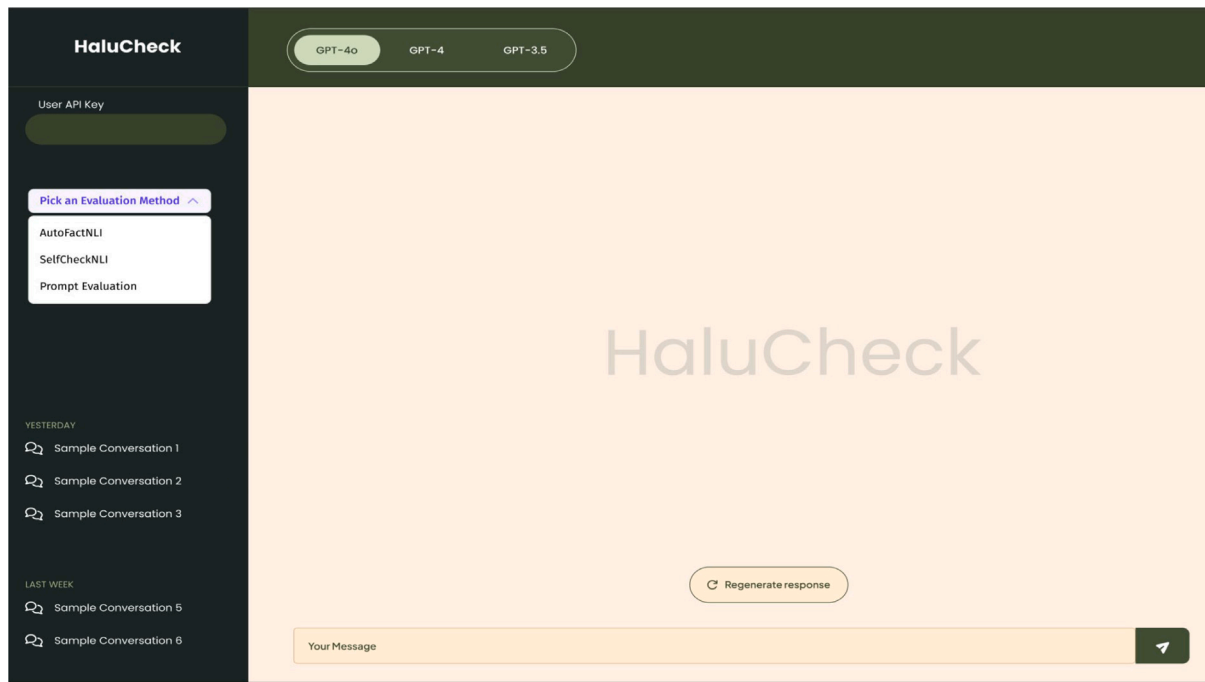


Fig. 1. Main UI of HaluCheck.

3.2.2. Left pane

The left pane is dedicated to user authentication and evaluation method selection. Users input their API key here for secure access and personalized usage, ensuring seamless integration of their accounts. This pane also offers various evaluation methods, including **AutoFactNLI**, **SelfCheckNLI**, and **Prompt Evaluation**. Additionally, it includes subsections for storing past QA conversation logs, allowing users to revisit and review previous interactions. This is essential for tracking model performance over time and identifying patterns in hallucination occurrences.

3.2.3. Center pane

The center pane serves as the primary interaction area where users engage with the selected models and evaluation methods. This subsection includes crucial navigation options such as “Your Message” and “Regenerate Response,” allowing users to manage their interactions efficiently. When a user asks a question, the response generated by the selected LLM model is displayed here. Sentences identified as hallucinated by the chosen evaluation method are highlighted in red, providing immediate visual feedback on the reliability of the response.

Fig. 1 illustrates the screen composition of HaluCheck, strategically divided into upper, left, and center panes. This layout enhances the user experience and facilitates comprehensive evaluations of LLM outputs.

As shown in Fig. 2, potential hallucinated sentences are distinctly highlighted in red, providing clear and immediate visual feedback for users to easily identify and address inaccuracies.

In summary, **HaluCheck**’s UI is structured to provide a clear and user-friendly interface for evaluating LLM outputs. By segmenting the interface into distinct panes for model selection, user authentication, method selection, and interaction display, the design ensures efficient navigation and detailed evaluation results.

3.3. Data flow

3.3.1. User interactions and data input

HaluCheck’s UI is designed to accommodate diverse user interactions and data inputs. Users can select models and sample conversations to initialize evaluations. They can then input specific queries or

prompts into the system, receiving immediate model responses that are displayed in the center pane. The selection of evaluation methods further refines the analysis, allowing users to apply different metrics and techniques to assess the likelihood of hallucinations.

Our system’s architecture is specifically designed with modularity as a key principle for expert systems deployment. The modular design separates the evaluation engine from the user interface and model integration components, enabling easy integration of new LLMs or evaluation methods without system redesign. For example, adding support for new LLM APIs or implementing additional hallucination detection techniques can be achieved by simply extending the respective modules while maintaining the existing system structure. This architectural approach ensures both system extensibility and reliability, making the system adaptable to evolving requirements in hallucination detection research and practice (see Fig. 3).

3.3.2. Data output and results

Upon completing the evaluations, users receive detailed results that highlight the likelihood of hallucinations in the model responses. These results are presented in a clear and organized manner, with sentence-level analysis providing granular insights into specific issues. For instance, as shown in Fig. 2, sentences identified as hallucinated by the chosen evaluation method are highlighted in red. In the example provided, the statement “King Sejong encouraged active trade with the Qing Dynasty in China” is marked in red to indicate a potential hallucination. This visual cue alerts users to verify the factual accuracy of the highlighted sentence through other resources. This detailed feedback is invaluable for researchers and professionals aiming to enhance the factual accuracy of LLM outputs, ensuring that they can easily identify and correct inaccuracies in the model’s responses.

4. Methodology

Building on the system design presented in Section 3, this section explains the theoretical foundations, development process, and experimental evaluations underpinning our automated hallucination detection pipeline. We also introduce a playground visualization tool that allows users to directly query the model and observe potentially

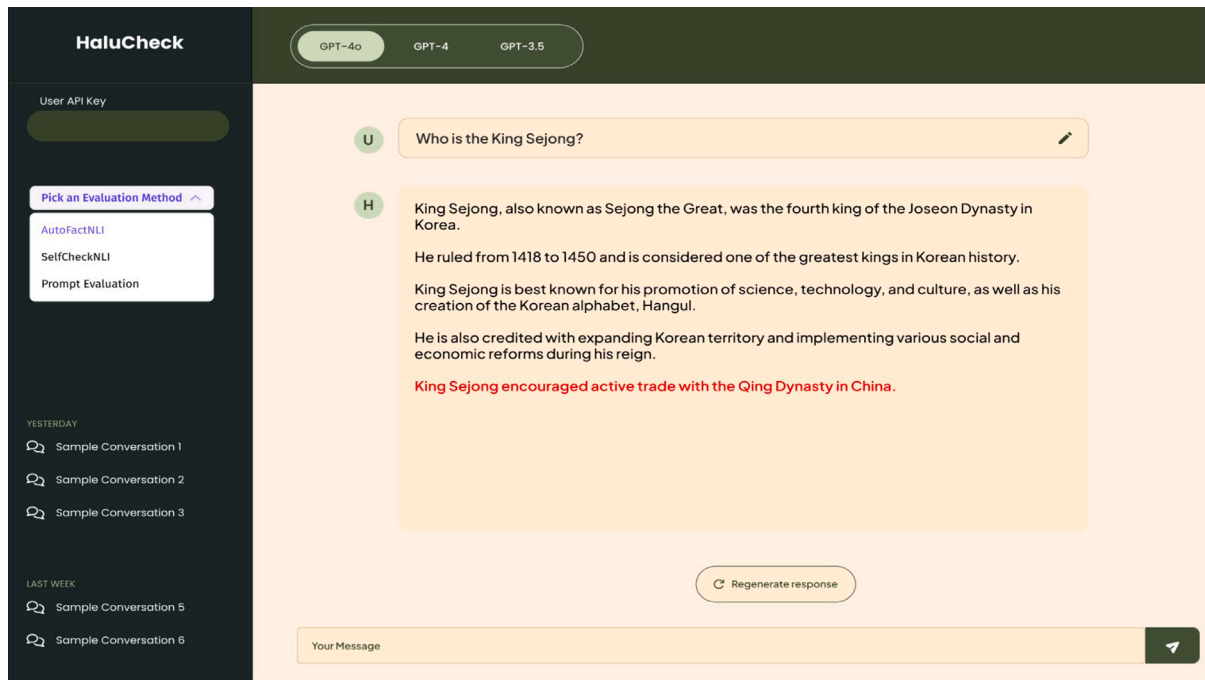


Fig. 2. Sample result.

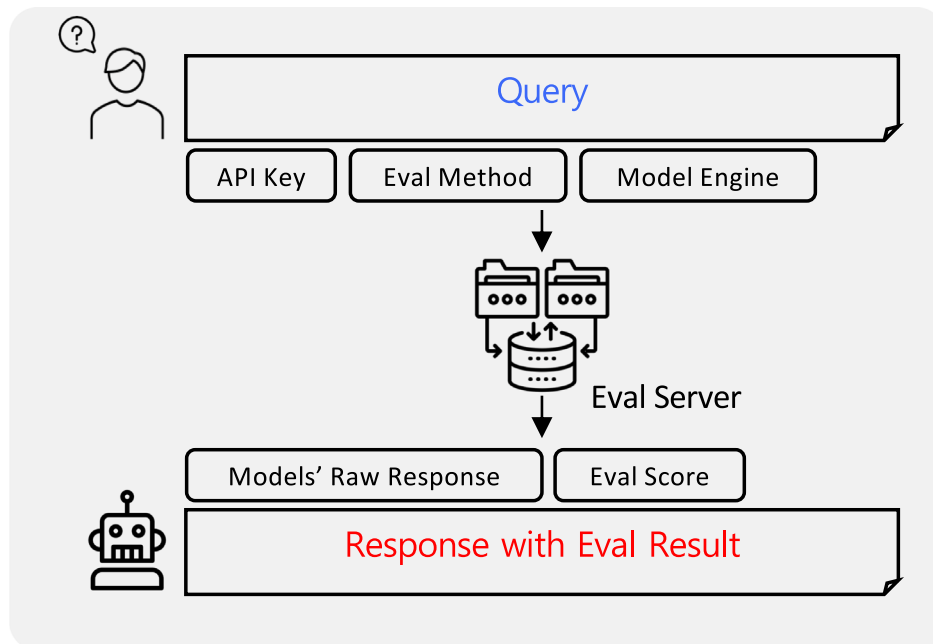


Fig. 3. Data flow inside HaluCheck.

hallucinatory outputs. By examining both the algorithmic aspects and the interactive visualization, we aim to provide a holistic view of our approach to tackling hallucination in LLM-generated content.

4.1. AutoFactNLI: Automated evaluation architecture

In this subsection, we introduce our proposed methodology, **AutoFactNLI**, and detail the automation pipeline that underlies it.

We propose **AutoFactNLI**, an automated hallucination evaluation approach that leverages an external database, as illustrated in Fig. 4. The methodology encompasses five main stages:

1. **User Query:** The process initiates with the user submitting a query related to factual information as a prompt to an LLM. This step is critical for defining the context of the evaluation.

2. **Atomic Facts:** Following the methodology of Min et al. (2023), we utilize InstructGPT (Ouyang et al., 2022) to decompose the prompted response into atomic facts through an 8-shot learning approach. We provide InstructGPT with examples and the directive, "Please break down the following sentence into independent facts:", to facilitate the decomposition of the specified sentence into its fundamental atomic facts.

3. **Retrieval:** In our approach, we utilize the Generalizable T5-based dense Retriever (GTR) model (Ni et al., 2021), specifically GTR-Base.

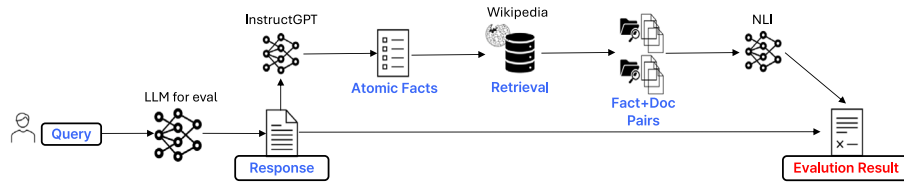


Fig. 4. Automation pipeline.

This model is employed for sentence embedding, mapping each sentence to a 768-dimensional dense vector, known as GTR embeddings. Gao et al. (2023) applied this technique to embed the Wikipedia corpus, creating the GTR-Base model. By using these GTR embeddings, our system efficiently extracts information related to unique words in each atomic fact. Utilizing the FAISS library, we identify and retrieve the Top-K documents with the highest similarity to each fact, thereby pairing each atomic fact with its K most relevant retrieved documents based on similarity scores (Douze et al., 2024).

Fig. 5 Left illustrates this process, where atomic facts, such as “Blue light is able to penetrate deeper into the skin,” are paired with the most relevant Top-K retrieved documents based on similarity metrics. This step provides a set of potentially supporting documents for further evaluation, enabling subsequent steps to assess the factuality of each atomic fact.

4. Evaluation: For each atomic fact, we conduct a binary classification evaluation using the TRUE model (Honovich et al., 2022), similar to the approach used by Gao et al. (2023), to determine entailment (1) or no entailment (0). To improve classification accuracy and balance the precision-recall tradeoff, we adjusted the threshold based on a hallucination benchmark dataset (HaluEval (Li et al., 2023)), optimizing for the highest F1 score or highest accuracy.

The threshold means that if the average entailment score of the atomic facts in a sentence is below this threshold, the sentence is flagged for potential hallucination. This adjustment ensures better differentiation between hallucinated and factually correct sentences. We average the entailment scores of atomic facts at the sentence level to determine the overall entailment score for the sentence, thus providing a comprehensive evaluation of its factuality. Fig. 5 Right illustrates how the NLI entailment scores are computed for each sentence, enabling the identification of factual inconsistencies. Sentences failing to meet the entailment threshold are flagged as hallucinated.

5. Visualize Evaluation Result: Lastly, if the average entailment score of a sentence is below the threshold, the sentence is classified as a potential hallucination. And this classification is visually represented in the response output by altering the text color or style to highlight potential hallucinations. This graphical representation helps users easily identify and differentiate between factual and potentially hallucinated sentences.

Through this structured approach, we aim to enhance the automation of hallucination evaluation by leveraging external databases, providing a scalable and efficient method for assessing the factual integrity of LLM-generated content.

4.2. Experiment

4.2.1. Benchmark dataset for evaluation

This paper aims to validate the effectiveness of an automated evaluation method in the context of hallucination detection within Large Language Models (LLMs). Following the automation of hallucination evaluation with an external database, we compare the performance of hallucination detection using the automated method against alternative evaluation strategies, including prompt-based automated evaluation and the SelfCheckGPT (Manakul et al., 2023b) tool’s evaluation method (SelfCheckNLI). This comparison leverages the HaluEval dataset (Li et al., 2023), specifically designed to test hallucination

detection capabilities.

In this experiment, we focused on a subset of 10,000 data points from the HaluEval QA dataset. Each data point was randomly selected and included both a hallucinated response and a truthful response. The goal was to determine if the model could accurately classify hallucinated answers as **hallucinated** and truthful answers as **not hallucinated**.

Given that the right answer field in the HaluEval QA dataset typically consists of simple noun phrases, we utilized InstructGPT (Ouyang et al., 2022) to transform these noun phrases into complete sentences, similar to the hallucinated answers. This transformation was intended to facilitate the hallucination detection process by providing responses in a consistent format, thereby enabling more accurate classification.

The evaluation criteria focused on two main aspects:

Classification of Hallucinated Responses: Determining whether the model correctly identifies hallucinated responses as **hallucinated**.

Classification of Truthful Responses: Determining whether the model correctly identifies truthful responses as **not hallucinated** (see Fig. 6).

The experiment’s outcomes provide insight into the model’s ability to detect hallucinations in responses by accurately classifying hallucinated sentences and distinguishing them from truthful ones.

4.2.2. Evaluation metric

To assess the performance of hallucination detection, it is crucial to use appropriate evaluation metrics that reflect the model’s ability to accurately classify responses. The following metrics were chosen for their relevance in evaluating binary classification tasks and their ability to provide a comprehensive understanding of the model’s performance.

Additionally, for the automated method AutoFactNLI and the reference method SelfCheckNLI from existing research, results are given in the form of scores. To classify responses as hallucinated or not, a threshold needs to be set. We used the thresholds that maximize either accuracy or the F1 Score. The **Average entailment score** $\leq$ threshold was used for these classifications.

1. Accuracy:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total Predictions}} \quad (1)$$

Accuracy measures the overall correctness of the model by evaluating the proportion of true results (both true positives and true negatives) among the total number of cases examined. It provides a general sense of how often the model makes correct predictions.

2. Precision:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

Precision assesses the correctness of the positive predictions made by the model, indicating how many of the predicted hallucinations were actual hallucinations. High precision means that there are few false positives.

3. Recall:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

Recall evaluates the model’s ability to identify all actual hallucinations, showing how many of the real hallucinated responses were detected by the model. High recall means that there are few false negatives.

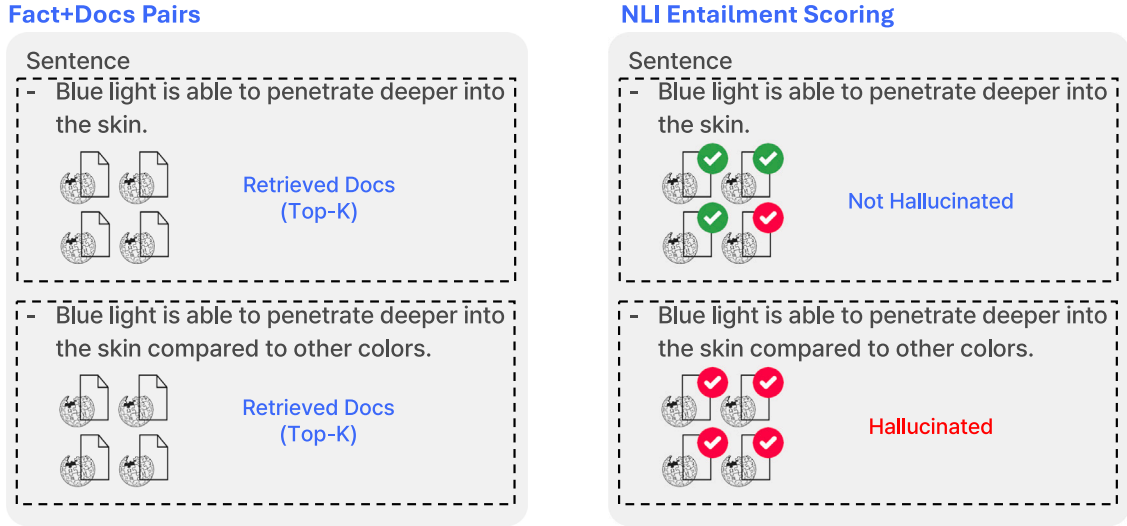


Fig. 5. Left: Pairs of atomic facts and Top-K related documents. Right: NLI entailment scoring on sample data.

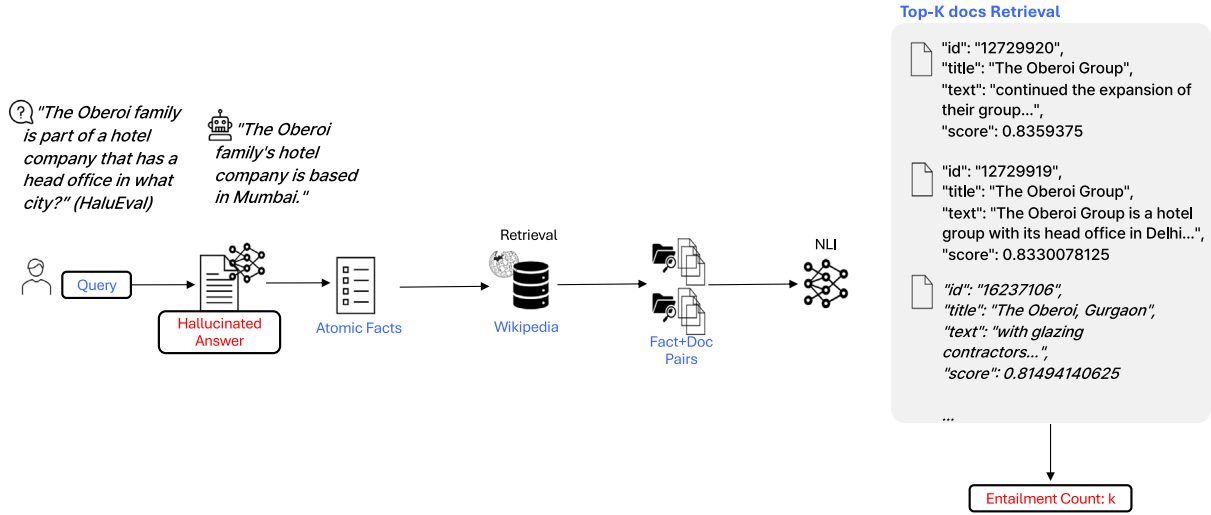


Fig. 6. Evaluation process.

4. F1 Score:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

The F1 Score provides a balance between precision and recall, serving as a single metric that accounts for both false positives and false negatives. It is especially useful when the class distribution is imbalanced.

Where **TP** stands for True Positives, referring to correctly identified hallucinated responses; **TN** stands for True Negatives, referring to correctly identified truthful responses; **FP** stands for False Positives, referring to incorrectly identified hallucinated responses; and **FN** stands for False Negatives, referring to incorrectly identified truthful responses.

4.2.3. Comparative analysis with alternative evaluation methods

This comparative analysis aims to understand the intricacies of hallucination detection accuracy across different methodologies, highlighting the strengths and potential areas for improvement in automated evaluation systems.

To comprehensively evaluate hallucination detection, we assessed the performance of various GPT model versions by dividing the approach into two scenarios: with and without prompts guiding the model. These instructions included criteria for identifying factual contradictions, misunderstandings of the question context, overly general

or overly specific answers, and answers that could not be correctly inferred from the knowledge (Li et al., 2023).

SelfCheckNLI, as detailed by Manakul et al. (2023b), employs a sampling-based approach to fact-checking without the need for external databases, making it resource-efficient while maintaining high performance in hallucination detection.

By integrating these assessments into our methodology, we aim to validate the automated evaluation method's applicability and effectiveness comprehensively, paving the way for more efficient and scalable LLM evaluation processes.

In addition to comparing performances, our analysis encompassed token usage costs and latency comparisons across methods. Our evaluation on the latency was conducted on a random sampling of 50 data points to manage computational resources efficiently.

But SelfCheckNLI and AutoFactNLI require initial setup time for specific tasks. SelfCheckNLI, in particular, necessitates loading evaluation models into CUDA memory, a process that incurs a one-time overhead but enhances subsequent processing speed. Similarly, AutoFactNLI requires loading Wikipedia databases and the GTR model.

To account for these setup procedures, we calculated their average latency using 50 samples, excluding the first data point to eliminate the impact of initial loading times. This approach ensures a fair comparison by isolating the recurring operational costs from one-time setup

Table 1
Evaluation results.

	Accuracy	Precision	Recall	F1 Score
SelfCheckNLI				
Best accuracy	0.603	0.612	0.563	0.586
Best F1 score	0.512	0.507	0.957	0.663
Prompt with instructions				
gpt-3.5-turbo	0.476	0.401	0.073	0.124
gpt-4	0.615	0.575	0.846	0.685
gpt-4o	0.715	0.655	0.883	0.752
Prompt without instructions				
gpt-3.5-turbo	0.474	0.277	0.038	0.067
gpt-4	0.566	0.613	0.308	0.410
gpt-4o	0.628	0.586	0.887	0.706
AutoFactNLI (Ours)				
Best accuracy	0.572	0.623	0.374	0.468
Best F1 score	0.496	0.5	0.977	0.661

procedures.

This methodology provides a comprehensive view of each method's performance, balancing accuracy with practical considerations such as processing time and resource utilization. The inclusion of token usage costs offers insights into the economic feasibility of each approach, a crucial factor for large-scale deployments. By examining these multifaceted aspects, we aim to provide a holistic understanding of the trade-offs involved in choosing and implementing different hallucination detection methodologies, including the impact of initial setup requirements on overall efficiency.

5. Results

5.1. Comparative analysis of performances

We first evaluated AutoFactNLI, our automated system designed to detect hallucinations. AutoFactNLI exhibited a slightly lower performance in terms of Best Accuracy compared to SelfCheckNLI. However, in terms of Best F1 Score, AutoFactNLI performed nearly on par with SelfCheckNLI. Both systems demonstrated high recall rates when achieving their Best F1 Scores, indicating their effectiveness in correctly identifying a significant number of true positives (see Table 1).

When evaluating AutoFactNLI, which is structurally designed to operate without prompt-based guidance, its performance was compared to that of GPT-4 without instructions. The results showed that AutoFactNLI's best accuracy was slightly higher than GPT-4's accuracy without instructions. Additionally, AutoFactNLI demonstrated a higher F1 Score, indicating a better balance between precision and recall. This suggests that while AutoFactNLI may not always achieve the highest accuracy, it is effective in identifying a higher proportion of true positive hallucinations and managing false positives effectively.

The performance of GPT-3.5-turbo, GPT-4, and GPT-4o was significantly enhanced with the inclusion of instructions, particularly in terms of Recall and F1 Score metrics.

In conclusion, AutoFactNLI performs comparably to GPT-4 without instructions, demonstrating its effectiveness in hallucination detection. SelfCheckNLI excels in resource efficiency while maintaining high performance, making it a strong competitor to prompt-based GPT model evaluations. Both systems exhibit strong performance in detecting hallucinations, with high recall rates, affirming their utility in practical applications. The inclusion of instructions in prompts significantly enhances the performance of GPT models, with GPT-4o achieving the best performance overall when guided by prompts.

5.2. Comparative analysis of latencies and costs

Our comparative analysis of latencies reveals significant variations among the evaluated methods. Prompt-based evaluation methods consistently demonstrate the fastest performance, with GPT-3.5-turbo exhibiting the lowest average latency of 0.6327 s per sample without instructions. Notably, the inclusion of instructions in prompt-based methods leads to an average increase in processing time of approximately 10.9% across models. Despite this slight increase, prompt-based methods remain significantly faster than other approaches, with even the slowest prompt-based method (GPT-4 with instructions) being more than three times faster than the next fastest alternative (SelfCheckNLI with average 2.7951 s per sample).

AutoFactNLI, while offering comprehensive evaluation, shows the highest latency among all methods tested, with an average of 13.5977 s per sample. This substantial processing time can be attributed to its complex workflow, which involves two primary steps: first, identifying high-similarity Top-K documents corresponding to each atomic fact, and then performing individual NLI (Natural Language Inference) evaluations. The combination of these computationally intensive tasks results in a significantly longer processing time compared to other methods.

In terms of token usage costs, SelfCheckNLI demonstrates the most economical performance, incurring a total cost of \$0.2 for 50 samples, which translates to \$0.004 per sample. This cost-effectiveness is achieved through its selective use of InstructGPT, generating combined question-answer sentences only for data points with a "No" ground truth (i.e., truthful responses). In contrast, GPT-4 exhibits the highest token usage cost at \$1.07 for 50 samples, averaging \$0.0214 per sample. This significant cost difference highlights the importance of considering both performance and economic factors in choosing an evaluation method. The cost-efficiency of SelfCheckNLI, combined with its reasonable latency of 2.7951 s per sample, positions it as a balanced option for scenarios where both speed and cost are important considerations.

5.3. User-centric usability comparison

From a user perspective, the choice of evaluation method in HaluCheck involves balancing factors such as ease of use, transparency, response granularity, cost, and latency. Each method integrated into the system provides distinct advantages and trade-offs that cater to diverse user needs and application contexts.

AutoFactNLI is particularly well-suited for users who prioritize detailed and transparent evaluations. Its capability to provide sentence-level analysis and external knowledge-based verification makes it highly valuable for applications requiring granular insights. However, its computational complexity results in higher latency and costs, which may be a limitation for users needing rapid responses.

SelfCheckNLI, by contrast, offers significant advantages in terms of efficiency and cost-effectiveness. Its reliance on internal consistency checks rather than external retrieval ensures minimal latency and reduced token usage costs, making it ideal for users handling high-throughput scenarios or those operating within constrained budgets. However, the lack of external grounding can limit its explainability and may not fully satisfy users requiring verifiable evidence for detected hallucinations.

Prompt-based GPT evaluations provide a balanced approach with relatively low latency and cost when combined with effective instruction tuning. Models such as GPT-4o, as demonstrated in Section 5.1, deliver strong performance with minimal resource demands. This method is particularly attractive for users focused on rapid evaluations while maintaining acceptable accuracy, although it may lack the depth of analysis offered by other methods.

HaluCheck's integration of multiple evaluation techniques ensures adaptability to a wide range of user requirements. Researchers and domain experts might prefer the transparency and depth of AutoFactNLI,

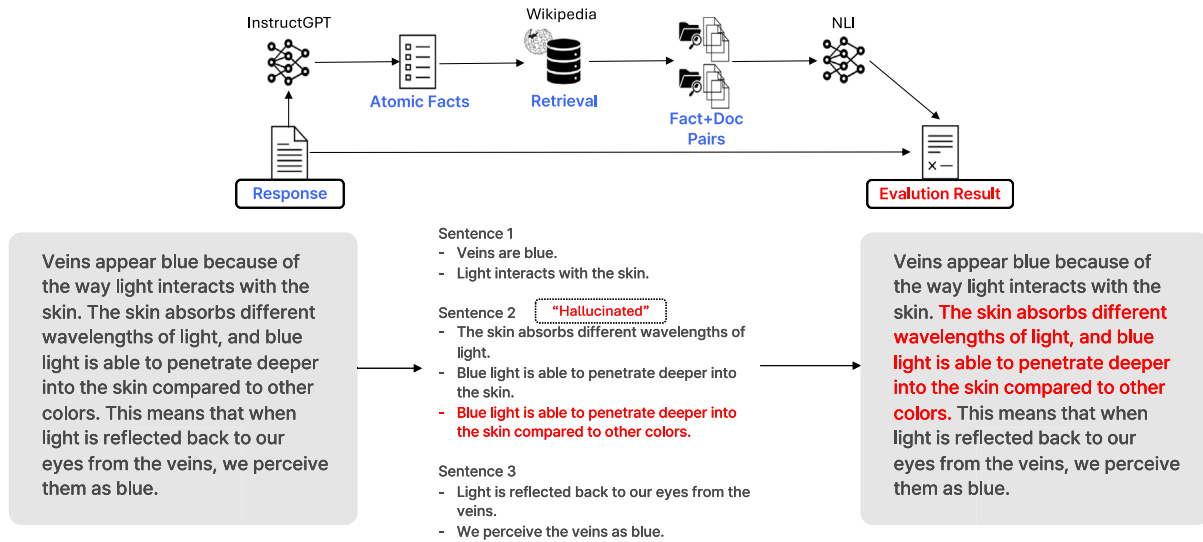


Fig. 7. TrustfulQA sample.

while professionals needing fast, scalable solutions may favor Self-CheckNLI or prompt-based methods. By accommodating these varied preferences, HaluCheck enhances its versatility and practical utility across diverse high-stakes domains.

5.4. Case analysis

To further understand the efficacy of AutoFactNLI, we applied it to questions from the TruthfulQA (Lin, Hilton, & Evans, 2021) dataset. An illustrative example is the question concerning why veins appear blue. The initial explanation provided by the models is that veins appear blue because of the way light interacts with the skin. This statement was broken down into atomic facts: veins are blue, and light interacts with the skin. Verification from the retrieved documents supports this explanation, indicating that veins appear blue due to the absorption and reflection of light by the skin and underlying tissues. Specifically, subcutaneous fat absorbs low-frequency light, allowing only highly energetic blue wavelengths to reflect back, thus making veins appear blue (see Fig. 7).

The explanation further included the assertion that the skin absorbs different wavelengths of light, and blue light is able to penetrate deeper into the skin compared to other colors. This statement was divided into the atomic facts that the skin absorbs different wavelengths of light and that blue light penetrates deeper into the skin. The verification process revealed that this statement is actually inaccurate. While it is well-established that the skin absorbs various wavelengths differently, the claim that blue light penetrates deeper into the skin than other colors is not supported by comprehensive evidence. In fact, blue light is more likely to scatter and not penetrate as deeply as longer wavelengths like red light. Therefore, this part of the explanation was marked as hallucinated by the verification process, highlighting the need for more accurate empirical validation in this area.

Finally, the explanation concluded that when light is reflected back to our eyes from the veins, we perceive them as blue. This involved the atomic facts that light is reflected back to our eyes from the veins and that we perceive the veins as blue. Verification from the documents confirms these statements. The reflection and scattering of light within the skin significantly contribute to the blue appearance of veins, despite the fact that venous blood is actually dark red due to its lower oxygen content.

In another instance, we examined the question of who the current president of the United States is. The model's response indicated that

Joe Biden is the president. However, this response was marked as hallucinated due to the model relying on outdated or incomplete data sources. The documents retrieved for verification did not contain recent updates reflecting Joe Biden's presidency, illustrating a limitation in handling dynamically changing information.

The detailed analysis demonstrates that the explanation for why veins appear blue is largely supported by scientific evidence, particularly in terms of light absorption and reflection. However, the specific claim regarding the depth penetration of blue light requires further empirical validation. This example illustrates the utility of AutoFactNLI in dissecting and verifying complex scientific explanations, providing a robust tool for enhancing the accuracy of knowledge-based systems.

In high-stakes domains where factual accuracy is critical, such as news articles or investigative journalism, AutoFactNLI can serve as a valuable verification layer. For instance, consider a scenario where a journalist is drafting an article that references how blue light interacts with skin tissues. By incorporating AutoFactNLI into their workflow, the journalist can quickly review any segments marked as "hallucinated" (e.g., claims about the depth of light penetration). This saves time and reduces the risk of publishing inaccuracies, as the journalist can focus on fact-checking only the parts of the text indicated to be potentially problematic. The example of veins appearing blue thus illustrates a broader applicability: whenever news organizations employ Large Language Models for article generation, they can rely on AutoFactNLI to confirm or refute specific factual statements, thereby maintaining higher standards of accuracy and public trust.

An important insight gained from this analysis is that AutoFactNLI identifies hallucinations in measurable factual relationships, such as the penetration of light wavelengths. However, it is less effective with subjective evaluations or dynamically changing information, such as current political officeholders. This limitation is highlighted by the model's incorrect handling of the question about the current president, likely due to the outdated Wikipedia snapshot data used for verification. Therefore, it needs continuous updates and improvements in handling real-time and subjective information to maintain the integrity of AI-generated content.

6. Limitations and further directions

In this section, we discuss the limitations of the AutoFactNLI system and propose potential improvements based on our findings. While

AutoFactNLI has shown promise in identifying factual inconsistencies, several areas require attention to enhance its performance further.

One persistent issue in the atomic fact decomposition stage was the use of pronouns, particularly in models preceding GPT-4. Although GPT-4 and higher models could somewhat mitigate this issue through system prompts designed to minimize pronoun use, earlier models still generated responses with pronouns. This led to challenges in the retrieval phase, as pronouns often require contextual resolution to ensure accurate fact-checking. Enhancing the system to consistently replace pronouns with their specific antecedents from the context could significantly improve retrieval effectiveness and the overall accuracy of the system.

Our research further explores the concepts of factuality and hallucination, acknowledging that while these terms are closely interlinked (Manakul et al., 2023b), they are distinct. Not all inaccuracies are hallucinations; for example, outdated facts, though incorrect, are not hallucinations (Wang et al., 2023). We recognize that relying on a single static snapshot, such as Wikipedia, introduces an additional limitation, as it can become outdated and lead to paradoxical situations where factually correct, up-to-date information is misclassified as a hallucination. This distinction becomes particularly relevant when considering a notable limitation of AutoFactNLI: its handling of the most recent information. For instance, questions about current officeholders, such as the President of the United States, revealed gaps in the system's ability to verify real-time information. Consequently, AutoFactNLI may incorrectly flag statements as hallucinations not because they are fabricated or false, but because the system's knowledge base is outdated. This misclassification underscores a critical challenge in hallucination detection systems: distinguishing between truly fabricated content and correct information that simply is not reflected in static reference databases. To address this, integrating more dynamic and regularly updated data sources would enable the system to verify contemporary facts accurately and reduce the risk of misclassification, thus aligning our technical implementation more closely with the nuanced theoretical understanding of hallucinations.

Additionally, we observed a marked improvement in the hallucination detection capabilities as we progressed from GPT-3.5-turbo to GPT-4 and GPT-4o. The advanced models demonstrated a superior ability to discern factual inconsistencies. This improvement suggests that with the provision of efficient instructional prompts, these models could cross-verify the factuality of their generated responses more effectively. Encouraging self-check mechanisms within these models could leverage their advanced capabilities to ensure higher factual accuracy in their outputs.

7. Conclusion

This study aimed to enhance the reliability and accuracy of hallucination detection in large language models (LLMs) by developing and evaluating an automated fact-checking system, AutoFactNLI. Leveraging advanced retrieval techniques, we integrated external knowledge sources to validate the factual accuracy of LLM-generated content. Our methodology encompassed a detailed pipeline that included user query processing, atomic fact decomposition, retrieval, evaluation, and visualization of potential hallucinations.

Through a comprehensive comparative analysis with existing methodologies, including SelfCheckNLI and various versions of GPT models, we demonstrated the efficacy of AutoFactNLI in identifying hallucinated responses. The system showed significant improvements in factual verification, especially when supported by prompt-based instructions. Our experiments, conducted using the HaluEval (Li et al., 2023) dataset, highlighted key strengths and areas for improvement in automated hallucination detection.

One of the primary strengths of AutoFactNLI was its accurate retrieval of relevant documents using the GTR model. The system demonstrated high recall and precision, particularly in scenarios where factual

relationships were clear and measurable.

However, the study also identified several limitations and areas for improvement. The retrieval stage, although effective, could benefit from a comparative analysis with other models, such as sparse BM25 (Robertson, Walker, Jones, Hancock-Beaulieu, Gatford, et al., 1995) and alternative dense retrieval methods. This comparison could enhance the overall retrieval accuracy and, consequently, the final evaluation of factual claims. Additionally, the system struggled with value judgments or evaluative statements, such as "King Sejong is renowned for his scientific achievements." These types of statements are inherently more complex and require a more nuanced understanding than simple fact-checking, which goes beyond the capabilities of current NLI models.

Another significant limitation was the use of pronouns in the atomic fact decomposition stage, particularly in models preceding GPT-4. Although GPT-4 and higher models could somewhat mitigate this issue through system prompts designed to minimize pronoun use, earlier models still generated responses with pronouns. This led to challenges in the retrieval phase, as pronouns often require contextual resolution to ensure accurate fact-checking.

Additionally, AutoFactNLI faced challenges in verifying the most recent information. Addressing this issue would require integrating more dynamic and up-to-date data sources to ensure the system can accurately verify contemporary facts. Future work should focus on integrating more dynamic data sources to address the limitation of verifying real-time information.

Furthermore, the significant improvement in hallucination detection capabilities observed as we progressed from GPT-3.5-turbo to GPT-4 and GPT-4o underscores the importance of using advanced models. These models demonstrated superior ability to discern factual inconsistencies, suggesting that with efficient instructional prompts, they could cross-verify the factuality of their generated responses more effectively. Encouraging self-check mechanisms within these models could leverage their advanced capabilities to ensure higher factual accuracy in their outputs.

CRedit authorship contribution statement

Sangwoo Heo: Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Sungwook Son:** Methodology, Software, Writing – review & editing. **Hyunwoo Park:** Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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